

AutoResearch for Traffic Congestion Prediction

Segment-relative features break a labeling-driven F1 ceiling

STAT 390 Capstone • Sherry Huang • Northwestern University

One-sentence claim:

Adding 4 road-specific speed features to a LightGBM model raises 30-min-ahead congestion F1 from 0.560 → 0.681 on validation (+21.6%) and 0.618 on held-out test, breaking a ceiling that 18 hyperparameter experiments could not cross.

From Topic to AutoResearch Contract

How the research question was formalized

The Research Question

- Can speed history alone predict congestion 30 min ahead?
- No real-time feeds, no external map data
- Binary classification per road segment
- Chicago Traffic Tracker: 786K rows, Sept 2023
- Metric: validation binary F1 (positive = congested)

AutoResearch Contract

- Modify ONLY src/model.py
- src/run.py FROZEN — split, label, metric immutable
- Keep if val F1 improves, revert if not
- Git commit every improvement; git checkout every discard
- Test set locked — opened exactly once at the end
- All 30 runs logged to experiments/results.csv

Data, Label, and Baseline

What was measured and where we started

rows

786,420

Chicago Traffic Tracker, Sept 2023

class ratio

3.49 : 1

77.7% not-congested vs 22.3% congested

threshold

~23 mph

30th percentile of training speeds

forecast horizon

30 min

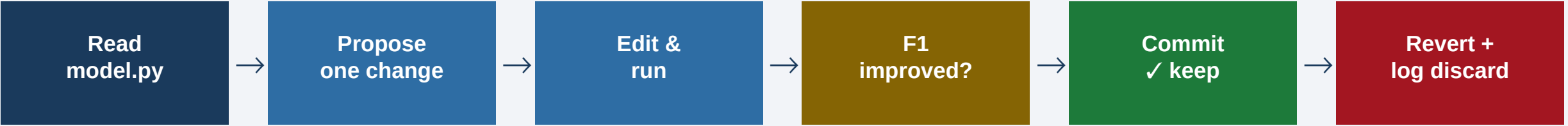
predict 3 steps ahead per segment

Key results — baseline vs. final model:

	F1	Precision	Recall	Notes
exp_001 Logistic Regression baseline	0.5598	0.7549	0.4448	Starting point
exp_030 Final LightGBM (validation)	0.6806	0.6352	0.7329	+21.6% gain
exp_030 Final LightGBM (TEST SET)	0.6183	0.5639	0.6843	Held-out, once

Loop Design and Experiment Trace

How the AutoResearch loop was structured and what it explored



30 experiments across 7 phases:

Phase	Experiments	Peak F1	Gain	Verdict
Model selection (LR → RF + balance + lags)	exp_001–007	0.6566	+0.097	RF wins
RF hyperparameter grid (depth, trees, ratio)	exp_008–013	0.6585	+0.002	Ceiling hit
Threshold sweep (0.35–0.50)	exp_014–018	0.6585	+0.000	Can't break it
Segment-relative features added	exp_019–021	0.6727	+0.014	CEILING BROKEN
Model comparison (RF/HGB/XGB/LightGBM)	exp_022–025	0.6780	+0.005	LightGBM wins
LightGBM HP tuning (leaves, subsample, n)	exp_026–030	0.6806	+0.003	Final locked

The Key Finding: A Labeling-Driven Ceiling

Why hyperparameter tuning hit a wall — and how we fixed it

The Problem

- Global threshold: speed < 23 mph → 'congested'
- But different roads have very different normal speeds
- Highway at 22 mph = normal rush-hour pace → labeled congested
- Side street at 20 mph = unusually fast → labeled not congested
- ~16.3% of training rows are mislabeled
- No model can learn from systematically wrong labels
- Hard ceiling at F1 $\approx$ 0.658 regardless of model or tuning

The Fix (no leakage)

- Compute per-segment stats from training data only:
- `segment_mean_speed` = mean(SPEED) per SEGMENT_ID
- `segment_std_speed` = std(SPEED) per SEGMENT_ID
- `speed_zscore` = (SPEED – mean) / std
- `speed_vs_seg_mean` = SPEED / mean
- Merge onto train+val+test by SEGMENT_ID (no leakage)
- Model now asks: 'slow for THIS road?'
- F1: 0.6585 → 0.6727 in one step (+0.014)

Final Result and Stability Check

exp_030: LightGBM tuned — test set opened exactly once

Validation F1

0.6806

model selection metric

TEST SET F1

0.6183

held-out, opened once

vs Baseline

+21.6%

F1 gain, validation

Stability std

±0.005

across 5 random seeds

Final Model (exp_030)

- LightGBM n_estimators=500, max_depth=6, lr=0.05
- num_leaves=31, min_child_samples=10
- colsample_bytree=0.8, subsample=0.8
- Undersample majority class 2:1, threshold=0.40
- 17 features incl. all 4 segment-relative features

Stability across 5 random seeds:

Seed	F1	Precision	Recall
42	0.671	0.595	0.769
0	0.668	0.600	0.755
123	0.674	0.600	0.768
7	0.682	0.636	0.736
99	0.674	0.621	0.737
Mean ± Std	0.674 ± 0.005	0.610 ± 0.016	0.753

Val→Test gap (−0.062): test period has 24.4% congestion vs 29.7% in validation — distribution shift in the later time window. No additional tuning was done.

What Worked vs. What Failed

Honest breakdown of all 30 experiments

✓ Worked

- **Segment-relative features** → +0.014 F1 (single biggest gain)
- **Undersample 2:1** → cleaner precision than `class_weight=balanced`
- **LightGBM over RF** → consistent +0.003 on identical setup
- **Subsampling regularization** → `colsample+subsample=0.8` → +0.003
- **Frozen evaluator** → all 30 comparisons valid and comparable
- **Git discipline** → zero lost experiments, zero accidental overrides

✗ Did Not Work

- **HistGradientBoosting** → failed both attempts (exp_005, exp_022)
- **Threshold below 0.40** → raises recall, always drops F1
- **RF hyperparameter tuning** → 6 experiments, total gain < 0.003
- **XGBoost** → competitive but LightGBM wins every comparison
- **Fixing the label definition** → requires unfreezing `run.py`
- **Single-seed evaluation** → lucky runs kept without knowing it

Reflection: Limits of the AutoResearch Loop

What did I learn about doing research with AI agents?

Agent excelled at

- Systematic bookkeeping — 30 exps, 0 errors
- Execution speed — hours, not weeks
- Identifying the labeling problem (16.3% disagreement)
- Clean model comparison (all variables held constant)

Agent struggled with

- No lookahead — committed to RF for 20 exps before LightGBM
- No repeat-on-change — HGB retry delayed 17 experiments
- Single-seed decisions — lucky runs not flagged
- Cannot diagnose why a ceiling exists — only measure it

Human judgment irreplaceable for

- Framing 'why is the ceiling there?' — not just measuring it
- Deciding when diminishing returns justify stopping
- Keeping run.py frozen — protects all comparisons retroactively
- Choosing which result is the story worth telling

Fundamental limits

- Congestion label still globally defined — residual label noise
- Val→Test gap (−0.062): later time window, different rate
- Single dataset, one month — generalizability unknown
- Greedy loop misses global optima by design

Key insight: the loop shifts the bottleneck from execution to diagnosis — asking the right question matters more than running more experiments.